PHYSIOL 578: HW 2.3

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1 You are studying two newly identified G-protein coupled Receptors (GPCRs): GPCR-A and GPCR-B. GPCR-A and GPCR-B bind and activate different heterotrimeric G-proteins. You treat cell lines that express either GPCR-A or GPCR-B with their respective agonist and observe that intracellular cAMP concentration changes as summarized in Table 1:

	Intracellular cAMP concentration	
	GPCR-A expressing cell line	**GPCR-B expressing cell line**
Agonist A for GPCR-A	Increase	No change
Agonist B for GPCR-B	No change	Decrease

Table 1: changes in intracellular cAMP concentrations

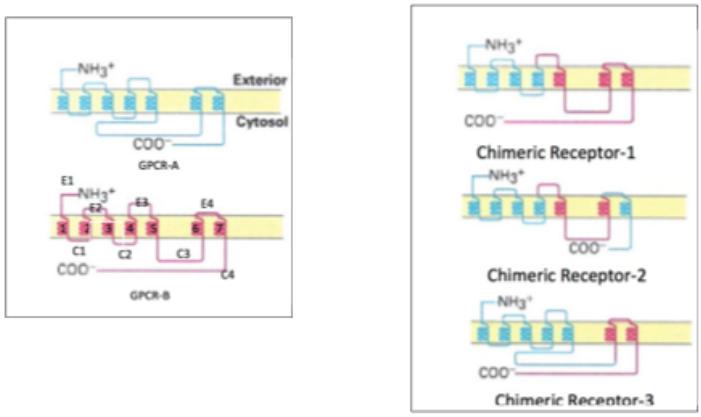
A. (1 point) Based on these observations and what you learned in PHYSIOL578, what type of heterotrimeric G-protein (Gs, Gi or Gq) does GPCR-A interact with to mediate its signaling? What about GPCR-B?

GPCR-A seems to interact with Gs since Gs activation stimulates adenylyl cyclase, increasing cAMP. On the other hand, since agonist for GPCR-B decreased cAMP concentration, it likely interacts with Gi.

U-M GPT was used to help draft and refine the answers.

B. (1 points) To investigate within GPCR-A and GPCR-B the agonist binding domain and the G protein-GPCR interacting domain, you generate 3 chimeric receptors as shown in Figure 1. You treat the 3 cell lines with either agonist A or B, and see the following changes in intracellular cAMP concentrations in Table 2. Based on these observations and what you learned in PHYSIOL578, which domains are *potentially* involved in agonist binding and which domains are involved in G protein-GPCR interaction? When describing the domain, please use E1, E2..E4 to describe extracellular domains; C1, C2...C4 to describe cytosolic domains; and helix 1 to 7 to describe transmembrane domains.

Figure 1: Chimeric Receptors



	cAMP		
	Chimeric Receptor-1 expressing cell line	**Chimeric Receptor-2 expressing cell line**	**Chimeric Receptor-3 expressing cell line**
Agonist A	No change	Decrease	No change
Agonist B	Decrease	No change	Increase

Table 2: changes in intracellular cAMP concentrations

Since chimeric receptor 1 did not see any activity from agonist A, we deduce that one of its domains from GPCR-B is responsible for binding: E3, E4, C3, C4, helices 5, 6, and 7. Since chimeric receptor 2 saw activity from agonist A, we deduce that one of its domains from GPCR-A is responsible for binding: E1, E2, E4, C1, C2, C4, helices 1, 2, 3, 4, 7. Since chimeric receptor 3 saw no activity from agonist A, we deduce that one of its domains from GPCR-B is responsible for binding: E4, C4, helices 6 and 7. Together, these leave E4, C4, and helix 7 as possible candidates. However, we know that the agonist must bind in extracellular space, so E4 and/or helix 7 must be the binding domain.

We can confirm the results with whether agonist B induced activity in the chimeric receptors.

We go through the same process for the domains involved in G-protein and GPCR interaction. Since chimeric receptor 2 saw a decrease in cAMP with agonist A, we deduce that one of its domains from GPCR-B is responsible for the interaction: E3, C3, and helices 5 and 6. Since chimeric receptor 1 saw a decrease in cAMP with agonist B, we deduce that one of its domains from GPCR-B is responsible for the interaction: E3, E4, C3, C4, helices 5, 6, and 7. Since chimeric receptor 3 saw an increase in cAMP with agonist B, we deduce that one of its domains from GPCR-A is responsible for the interaction: E1, E2, E3, C1, C2, C3, helices 1 through 5. Together, these leave E3, C3, and helix 5 as the only possible domain. However, we know that the G-protein and GPCR interact in the cytosol, so C3 and/or helix 5 must be the interaction domain.

C. (1 points) What effect will adding pertussis toxin have on cellular cAMP levels when added in addition to agonist A or to agonist B for all 5 cell lines (cells expressing GPCR-A, GPCR-B, Chimeric Receptor-1, 2, or 3). Feel free to use a table format.

Pertussis toxin only affects Gi by keeping it in inactive state. This means whichever receptors have GPCR-B interaction domain and decrease cAMP concentration by activating Gi will see an increase in cAMP since the inhibition of adenylyl cylcase is removed.

	cAMP				
	GPCR-A expressing cell line	**GPCR-B expressing cell line**	**Chimeric Receptor-1 expressing cell line**	**Chimeric Receptor-2 expressing cell line**	**Chimeric Receptor-3 expressing cell line**
Agonist A	No effect due to Gs	No effect due to inactive GPCR	No effect due to inactive GPCR	Increase	No effect due to inactive GPCR
Agonist B	No effect due to inactive GPCR	Increase	Increase	No effect due to inactive GPCR	No effect due to Gs